

Λ CDM is being contested as the latest results of evolving dark energy from DESI and various tensions (namely H_0 and S_8) test our understanding of the Universe. Large-scale weak lensing galaxy surveys offer constraints on cosmological parameters in various redshifts along with independent systematics compared to spectroscopy and CMB studies. While there are a variety of probes that galaxy surveys can use, I am particularly excited by galaxy clusters as they can give us strong constraints on $\sigma_8(z)$, $\Omega_M(z)$, and $f_{\text{gas}}(z)$ all of which are fundamental to understand the growth of structure in the Universe.

Galaxy clusters are some of the largest gravitationally bound structures in the Universe making them ripe for weak lensing studies. Furthermore, they are ideal science validation targets for early Rubin, Roman, and Euclid data as they have relatively light shear and redshift calibration requirements. My ultimate goal is to use these instruments to study galaxy clusters via weak lensing and constrain the redshift evolution of cosmological parameters. Over the next few years, I will perform weak lensing mass measurements of clusters in that can directly feed into a cluster cosmology study and start to measure any redshift dependence of derived parameters. To that end, I plan to validate the science pipelines by measuring any inherent biases and work on novel methods of mitigating weak lensing systematics.

Previous and Current Research The weak lensing systematic I have focused on during my PhD is **unrecognized blends** which are objects (typically galaxies) that are mistakenly identified as a single entity due to severe amounts of blending (see Fig. 1 for an example). The repercussions of unrecognized blends via detection bias alone have been studied previously, showing shifts in cosmology parameters at the level of a few percent [**ben`clustering, blends`2011, kidsblends`2025**], and an early study including the impact on shape and flux measurements found up to a 2σ shift in S_8 when conducting a cosmic shear analysis [**erfan**].

Typically, unrecognized blends are dealt with by measuring bias corrections from expensive image simulations that are best suited for cosmic shear studies. Cluster weak lensing suffers from larger shears and more blending, limiting the applicability of this calibration technique. Using the calibrations from these simulations may be inaccurate for cluster lensing and assume that all relevant systematics are included in the simulations. Directly identifying unrecognized blends in the data is an appealing solution that I pursued.

Blending In [**unrec`liang**], we showed that unrecognized blends can be readily detected at the catalog level using machine learning techniques to assign a blend probability based on measured fluxes. While we showed that it is possible to remove unrecognized blends from the catalog, it was unclear if such a removal would ultimately improve our cosmology. I investigated this by studying the impact of our blend removal method on shear calibration using the AnaCal shear framework. We showed that not only does the bias reduce, **the bias can be almost totally removed** with a 30-40% sample cut [**unrec`shear**]. Given that LSST will be systematics limited, a 30% cut to remove a source bias may be worthwhile

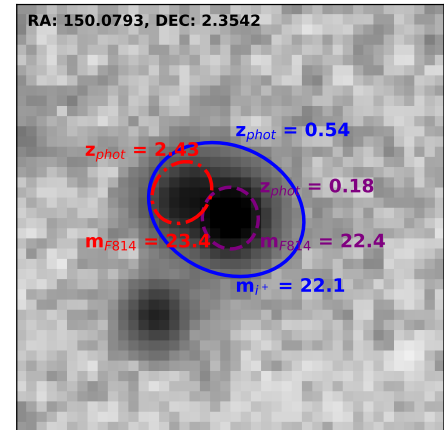


Figure 1: Example unrecognized blend in the COSMOS field with observed magnitudes and estimated photometric redshifts attached. Dark blue is the object as detected and measured by Subaru, dashed-dotted red and purple are detections and measurements from HST.

Clusters Outside of blending I have lead efforts to produce some of the first weak lensing results with the Vera Rubin Observatory by studying the galaxy cluster Abell 360 (Fig. 2). For that work I specifically worked on the source sample selection, photo- z validation with DESI, shear measurement with shapeHSM and AnaCal, and cluster mass estimation. As the scope of this project grew, my responsibilities also extended beyond purely scientific as I took on the role of data manager, daily meeting coordinator, and writing coordinator. Along with being my first foray into clusters, this was my first time leading a large collaborative effort.

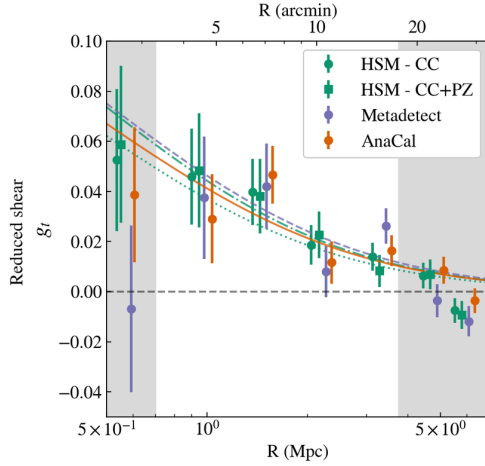


Figure 2: Tangential shear signal measured around Abell 360 using a variety of shape measurements and source selection. [paper in prep.]

RAIL, and Metadetect.

Research Proposal During my postdoc, I plan to focus on the early data from Roman, Rubin, and Euclid primarily looking into cluster weak lensing. I will validate and measure cluster weak lensing profiles along with working on reducing weak lensing systematics.

Goal 1: Cluster measurement validation for Roman and Rubin

While the shear and redshift calibration requirements are not as stringent as cosmic shear, precise ensemble measurements require an estimate of biases from both statistics and systematics. For clusters this typically is done with N-body or hydrodynamic simulations that can capture any physics-induced bias (miscentering of optical clusters due to triaxiality), but is expensive and can ignore biases coming from image processing. I propose to work on both direct image generation and synthetic source injection (SSI) to validate shear and redshift measurements coming from science pipelines.

Direct image generation has been the standard technique to calibrate any biases in cosmic shear [macrann, hsc] as it is able to directly test for systematic biases inherent in the tools and algorithms being used. There are a number of improvements we need to make to these image simulations to properly test clusters, including varying background redshifts, introducing cluster members, intra-cluster light, and cluster triaxiality. Introducing these systematics into our simulations sequentially will allow us to understand the bias introduced by each feature to inform future work on mitigation strategies.

While image generation is a powerful tool, it relies on correctly understanding and implementing all the systematics that exist in data. Alternatively, SSI allows us to characterize the existing

Following my efforts on Abell 360, I have been working on testing the combination of AnaCal and RAIL which produce a unified shear photo- z catalog on clusters in the Euclid Deep Field South field [publication in prep.] In this work we have **measured tangential shear signals around clusters that were undetectable in the base Rubin catalog**, highlighting the power of the combination of AnaCal and RAIL. This is one of the first examples of using these tools on observed data and gives a baseline for subsequent data releases by Rubin.

My previous experiences with both clusters and weak lensing systematics have equipped me to lead efforts on measuring the tangential shear signal around clusters using modern cosmology measurement tools like AnaCal,

data and test the measurement pipeline directly on synthetic sources. In the Abell 360 work, this technique was used to understand the effect of various cuts on the source sample and validate star-galaxy separation along with flux measurement. Repeating this work for Roman and Rubin clusters on a larger scale will allow us to validate measurements made on the real data.

Performing both image simulation and SSI gives us an understanding of what causes biases and the total level of bias present in the data. Improving on the existing image simulation tools and developing the pipelines to measure biases via SSI are necessary steps for future data releases with more stringent systematics requirements. My experiences with image simulation to validate blend removal and Abell 360 cluster measurements make cluster validation a natural next step that I am qualified to work on and lead.

Goal 2: Cluster measurements with Rubin in early data

With a validated pipeline in hand, we will have confident (or at least properly weighted) cluster weak lensing masses. For Rubin Data Preview 2 there are ~ 100 X-Ray detected clusters by eRosita and ~ 400 SZ detected clusters by ACT, which are more than enough to produce a LSST mass calibration for those external samples. The previous ACT calibration uses a similar number of clusters [citation], so it is possible to produce competitive results using this early data alone. Producing cosmology contours from calibration constitutes another project that would require estimating a covariance, modeling shear profiles (either hierarchically or stacked), redshift distribution estimation, and running cosmology chains, all of which are quite new to me. I am excited to learn and build towards a tomographic understanding of the cosmos.

Goal 3: Mitigate weak lensing systematics

I will continue to work on mitigating weak lensing systematics, especially blending. While my previous work on blending has focused on removing these extremely blended objects, it may be possible to model the effects of blending for better mitigation without incurring a statistical cost. Using the machine learning probability from [unrec'liang] as a prior, we can treat certain objects as a superposition of two galaxies which can lead to more accurate posteriors.

For redshift distributions, the self-organizing map (SOM) which maps measured fluxes to a 2D surface, is an effective tool for estimating ensemble distributions $n(z)$ but can also be extended to account for blends as outlined in Fig 3. Instead of asking for the single SOM cell that best maps the measured fluxes, a second “blending SOM” can be made which can account for the possible inputs leading to a more accurate redshift probability distribution function. Similarly for shears, accounting for cross-redshift blends in the measurement step may improve deblending and decrease shape noise.

Ultimately, making use of space based imaging is one of the best solutions to unrecognized blends as the smaller PSF reduces the amount of overlap from neighboring sources making **joint processing an extremely powerful solution**. While there are strong solutions to best combine photometry, there is also work to be done on combining shape measurements to maximize weak lensing signal. During my postdoc, I plan to contribute to the on-going efforts to enable joint-processing for both Metadetect and AnaCal. This will mostly rely on simulated datasets, like the existing Roman-Rubin simulations, but can be tested on data using deep field overlap of clusters fields.

This is an extremely exciting time to be doing cosmology and I have been at the forefront working with some of the first images taken by Rubin. I want to build on my experiences from my

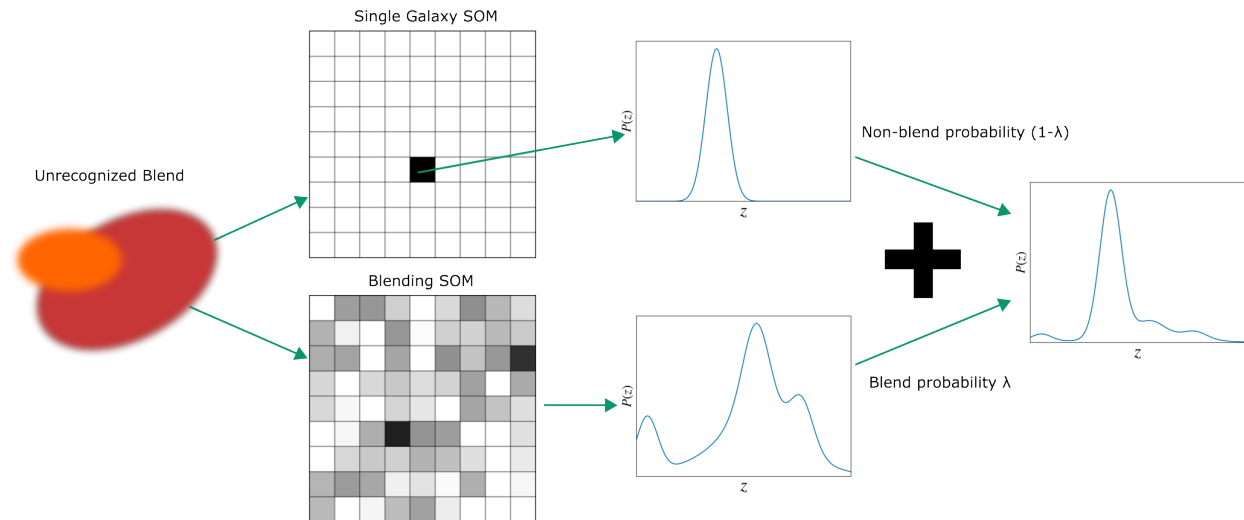


Figure 3: Example of modeling blends when producing redshift estimates via a SOM. Instead of only using the cell of the best fit, a weighted SOM can be made from synthetic blends that can be used to produce a more accurate PDF.

PhD by working on weak lensing systematics and clusters so that we can better probe the universe tomographically.